

MORPH Attention — Context Coverage at Scale

two parallel branches over one causal context — CSA (sparse) and HCA (dense) on alternating layers, plus the local window

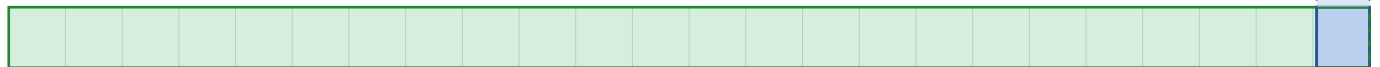
Even layers • CSA branch



sparse: top-128 of $S/4$ fine (4-tok) blocks — selected *anywhere* in $[0, i]$ (data-dependent)

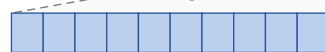
*one branch per layer — CSA (even) and HCA (odd) **alternate**, never run together; the window runs in both*

Odd layers • HCA branch



dense: all $S/128$ coarse (128-tok) blocks attended

(128 tok $\lll$ full width — enlarged here; axis not to scale)



local window: last $w=128$ tokens, attended **per-token** (in *both* branches)

1 HCA cell (128 tok)

32 CSA cells (4 tok each)

Boundaries coincide (1 HCA cell = 32 CSA cells, since $128=32 \times 4$ and both tile from token 0) — *but* CSA and HCA pool the raw tokens **independently** (separate compressors, alternating layers). Neither is built from the other; this is shared coverage, not a computed hierarchy.

CSA: sparse selected fine block

HCA: dense coarse block

window: per-token local (both branches)

At $S=1\text{M}$: HCA attends all $S/128 = 8,192$ coarse blocks (dense); CSA keeps top-128 of $S/4 = 262,144$ fine blocks (sparse, $\approx 0.05\%$); window = 128 tokens. Block sizes symbolic in S ($n_{\text{blk}}=S/m$, computed live).

Invariants: CSA and HCA *alternate by layer* (even $\rightarrow$ CSA, odd $\rightarrow$ HCA) — the two lanes never co-occur; the combined picture is what the *stack* sees. The **window** is always on. All attention is causal $[0, i]$.